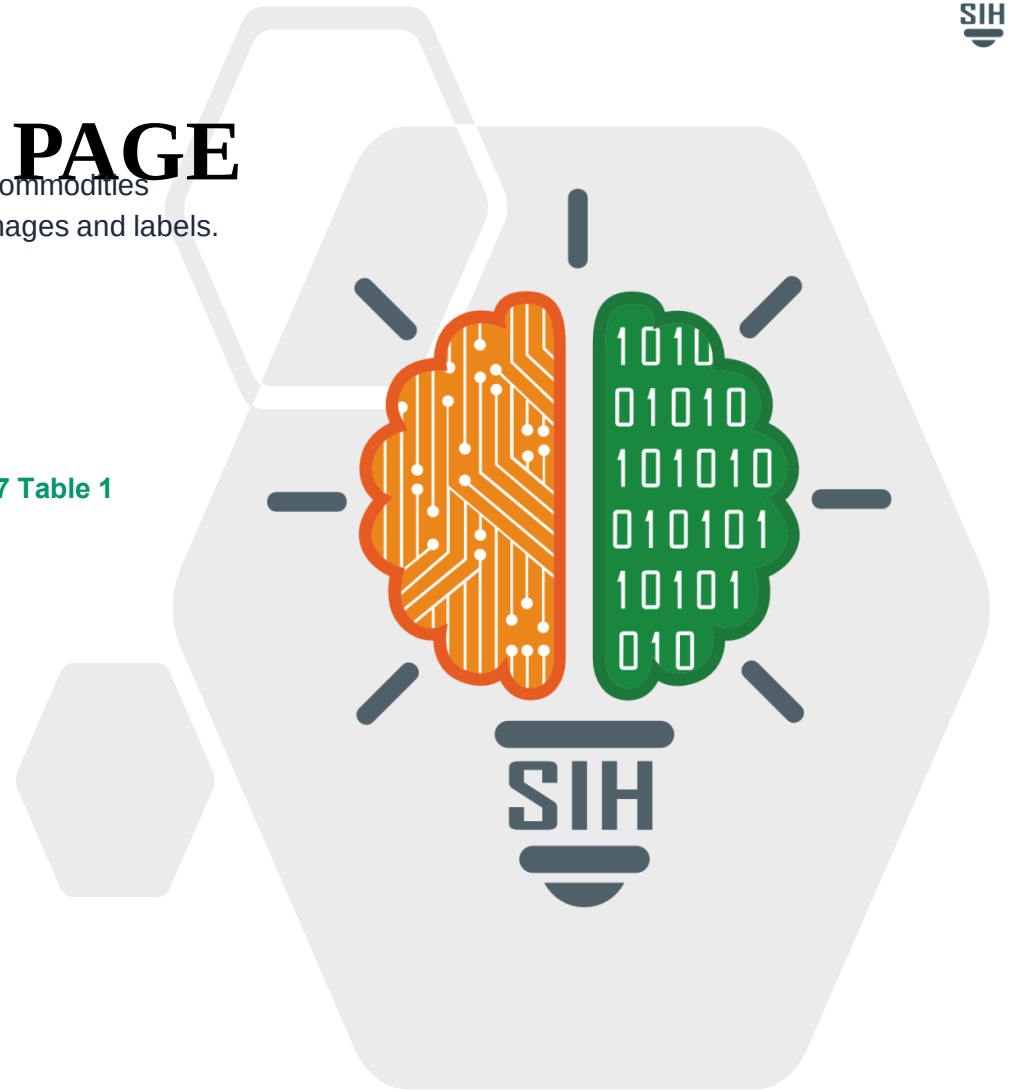
SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID —** 26034
- **Problem Statement Title —** Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.
- **Theme —** Miscellaneous **| PS Category —** Software
- **Team Name —** TECH SETU **| Product Name —** CompliScan
- **Live Working Prototype —** https://prototype-complisca.vercel.app/
- **GitHub Repository —** https://github.com/mufaddalbatti53/prototype-complisca
- **Deployment Status —** 100% Operational & Deployed Live (IST Synced • Rule 6 & 7 Table 1 Calibrated)



TECH SETU

COMPLISCAN — AUTOMATED PACKAGING COMPLIANCE ENGINE

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PROPOSED SOLUTION

CompliScan: Automated Packaging Compliance Engine

Snap a package. Get an official statutory compliance verdict in ~8–10 seconds.

• Multi-Channel Ingestion: Audits retail goods via Smartphone Camera, high-res photo upload, or quick-commerce listing URLs (Amazon, Blinkit, Zepto, Flipkart).

• Full Rule 6(1) Verification: Audits MRP, Net Quantity, Mfg/Exp Date, Unit Sale Price (USP), Consumer Care helplines, and Manufacturer/Packer addresses.

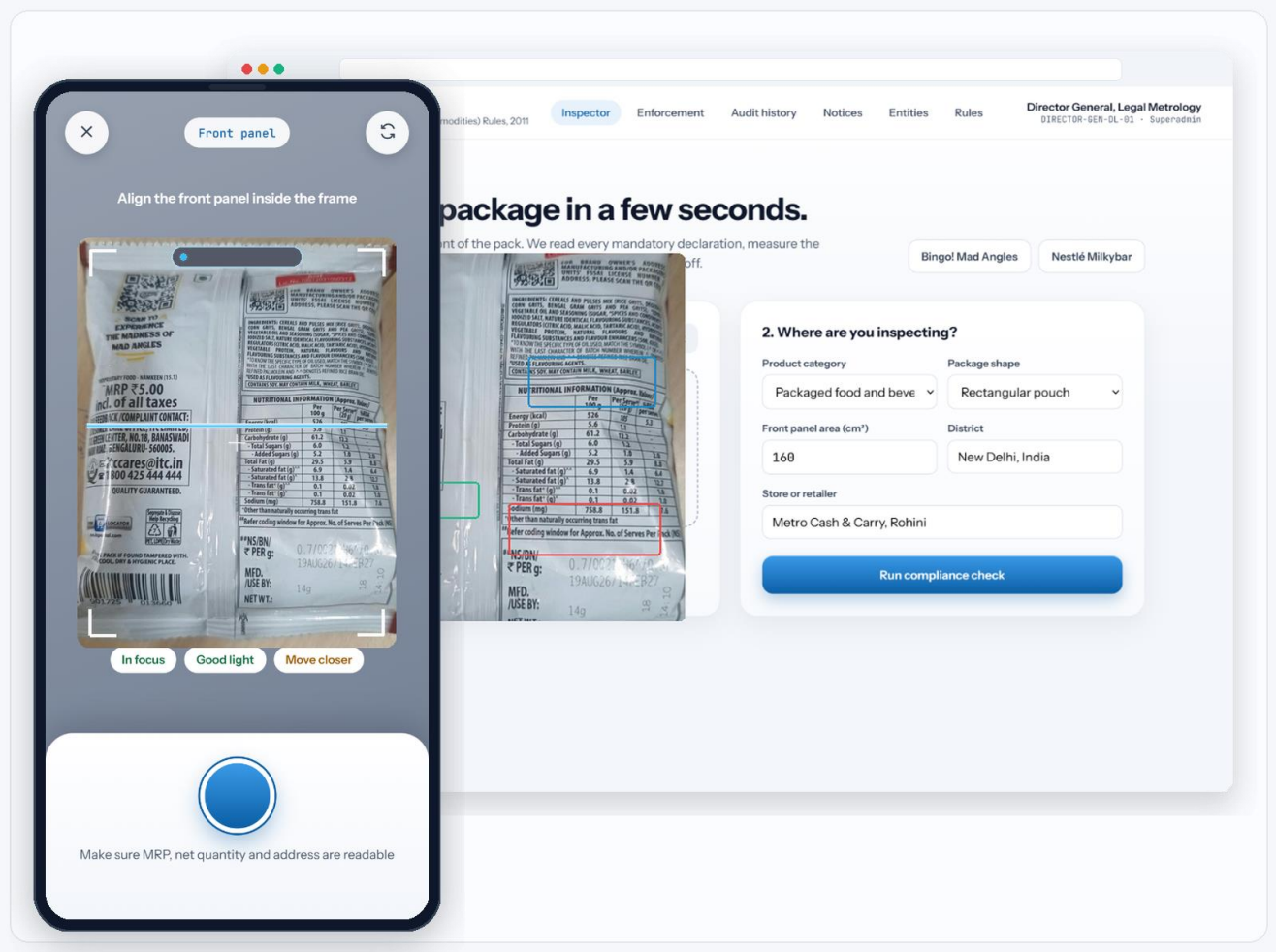
• Centralized Officer Workbench: Real-time telemetry, digital evidence locker, Form 1 Seizure Notices, and 4-stage statutory aging buckets.

WHY WE ARE DIFFERENT — WE CHECK LEGAL RULES, NOT JUST TEXT

• Rule 7 Table 1 Font Height: Computes Principal Display Panel (PDP) area in cm² and measures text height down to sub-millimeter precision against mandatory thresholds (1.0mm - 6.0mm).

• Mandatory Phrasing & Clauses: Validates 'MRP incl. of all taxes', 'Unit Sale Price', and standardized metric units under LMPC 2011.

• Automated Legal Compounding: Pre-fills Form 1 Notice of Seizure and calculates compounding fines under Section 18 / 36 of Legal Metrology Act, 2009.



Live Working Prototype: Real-Time Viewfinder Scan + Legal Metrology Audit Workbench (~8–10s Pipeline) — https://prototype-complisca.vercel.app/

TECH SETU

TECHNICAL APPROACH

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HOW IT WORKS — FROM PHOTO TO RESULT

1. CAPTURE	2. API GATEWAY	3. PRE-PROCESS	4. VISION OCR	5. LEGAL ENGINE	6. NOTICE & REPO
React 18 + Vite PWA Camera with framing guide, photo upload, or URL	**FastAPI Asynchronous** Uvicorn ASGI, JWT auth, strict IST serialization	**OpenCV (cv2) Engine** Perspective deskewing, contrast & DPI scaling	**Gemini Flash-Lite** + PaddleOCR engine, dot-matrix text reading	**LMPC Rules 2011** Rule 6, Rule 7 Table 1, Sec 18/36 compounding	**ReportLab PDF Notice** Form 1 seizure memo, 4-stage aging buckets

TECHNOLOGY STACK

FRONTEND	BACKEND	DATABASE	THE BRAIN	HOSTING
React.js + Vite Tailwind CSS	Python + FastAPI REST APIs	Supabase (Postgres + Storage + Auth)	Google Gemini Flash + PaddleOCR	Vercel (web) Render (API)

METHODOLOGY & PRODUCTION IMPLEMENTATION

- Microservices Architecture: Decoupled client-server design scaling horizontally with zero state bottlenecks.
- Configurable Statutory Rule Matrix: Legal rules stored as dynamic parameters—updates instantly without redeploying code.
- Multi-Stage Resilient Parser: Built-in json-repair & auto-healing eliminates errors on curved, reflective, or dot-matrix packaging.
- **Working Prototype Status:** 100% Operational & Deployed Live — https://prototype-complisca.vercel.app/ | **GitHub:** https://github.com/mufaddalbatti53/prototype-complisca

TECH SETU

FEASIBILITY AND VIABILITY

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~10 s	< ₹0.05	96%+	100%
End-to-End Pipeline	**Cost per Audit Scan**	**Field Compliance Accuracy**	**Digitally Sealed Records**
Photo to Form 1 Notice	Optimized token pipeline	On Indian retail packaging	Tamper-proof evidence locker

Production-verified benchmarks tracked continuously on our live enforcement dashboard.

FEASIBILITY & VIABILITY

- Technical: Production-ready FastAPI, OpenCV, and Gemini 3.1 Flash-Lite scale seamlessly to thousands of concurrent scans.
- Cost Efficiency: Near-zero operational overhead (<₹0.05/scan) via optimized flash-lite vision tokens and edge caching.
- Zero-Hardware Dependency: Field officers inspect packages in ~10s via any standard smartphone browser (PWA).
- Statutory Grounding: Every finding cites exact clauses under Legal Metrology Act, 2009 & LMPC Rules, 2011.

CHALLENGES & STRATEGIC SOLUTIONS

- Curved & Reflective Pouches → OpenCV homography deskewing & multi-panel scanning reconstruct distorted labels.
- Dot-Matrix Stamped Dates/MRP → Resilient OCR parsing handles faint, low-contrast dot-matrix stamping accurately.
- Multilingual Indian Packaging → Dual OCR & Multimodal Vision parses bilingual Hindi, English, and regional text.
- Low-Bandwidth Rural Areas → Lightweight PWA payload (<380 KB gzip) enables offline capture and reliable sync.

HUMAN-IN-THE-LOOP ADJUDICATION (OFFICER OVERSIGHT)

Officers retain complete adjudicatory authority. The AI serves as an enforcement copilot, not the final judge.

1. Confidence Guardrail: AI flags low-confidence readings (<85%) directly for officer verification.
2. Visual Evidence Overlay: Officer inspects bounding boxes, extracted strings, and physical packaging simultaneously.
3. 1-Click Rectification: Officer can accept, edit, or dismiss findings; every action is logged in the audit trail.
4. Court-Admissible Proof: Final Form 1 Notice contains verified officer endorsement and digital hash.

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IMPACT AND BENEFITS

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MANUAL INSPECTION VS COMPLISCAN

MANUAL FIELD INSPECTION (CURRENT STATUS)

15–20 minutes per package

- Visual ruler estimation — font height errors are routinely missed.
- Handwritten show-cause notices — heavy clerical and legal burden.
- Paper files — difficult to track repeat offenders or compile evidence.
- Frequent legal challenges due to lack of objective digital proof.

WITH COMPLISCAN ENFORCEMENT ENGINE

~8–10 seconds per package (>98% time reduction)

- Sub-millimeter digital font measurement against Rule 7 Table 1.
- 1-Click automated Form 1 Seizure Notice generation (PDF).
- Centralized evidence locker with 4-stage aging buckets (0-60+ days).
- Court-admissible timestamped audit logs (IST UTC+05:30).

LEGAL METROLOGY OFFICERS & DoCA

10x higher inspection throughput. Automated Form 1 memo generation, repeat offender tracking, and district-wide seizure analytics.

CONSUMER PROTECTION & RIGHTS

Eliminates deceptive packaging, unfair overcharging above MRP, misleading quantity stamps, and missing consumer grievance contacts.

MANUFACTURERS & E-COMMERCE

Pre-dispatch digital compliance auditing prevents expensive packaging seizures, market recalls, and Section 36 compounding fines.

Social: Restores consumer trust • Economic: Reduces compounding litigation & enforcement costs • Governance: Transparent data for DoCA • Environment: Paperless digital notices reduce waste

TECH SETU

RESEARCH AND REFERENCES

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MAIN SOURCES AND TOOLS WE USED

Legal Metrology Act, 2009 & LMPC Rules, 2011

Statutory benchmark for Section 15 (Seizure), Section 18 (Declarations), Section 36 (Penalties), and Rule 7 Table 1 (Font Height).

Ministry of Consumer Affairs, Food & Public Distribution — consumeraffairs.gov.in

Google Gemini 3.1 Flash-Lite Multimodal Vision API

High-throughput visual character recognition on curved, dot-matrix, and low-contrast consumer product packaging.

Sub-second Multimodal Inference Benchmarks — ai.google.dev

Department of Consumer Affairs (DoCA) & E-Commerce Rules

Guidelines for standard retail packaging, consumer redressal helplines, and LMPC (Amendment) Rules 2021 for digital marketplaces.

National Consumer Helpline (NCH) Standards — consumeraffairs.nic.in

FastAPI Asynchronous Framework & Computer Vision

High-performance ASGI microservices with OpenCV image deskewing, Pydantic v2 data validation, and ReportLab legal memo PDF engine.

Asynchronous Microservice Architecture — fastapi.tiangolo.com

PaddleOCR Multilingual Deep Learning Toolkit

Ultra-lightweight OCR architecture for text localization, angle rectification, and multi-script character recognition.

Bilingual English & Devanagari Benchmarks — github.com/PaddlePaddle/PaddleOCR

Supabase Cloud Database & Storage Infrastructure

Enterprise PostgreSQL database, row-level security (RLS), tamper-evident audit logs, and encrypted digital evidence locker.

Cloud Relational Architecture — supabase.com/docs